

NOVEL PREP / SAT MATHEMATICS

Unit 2: Advanced Math

CB Bank

Step-by-Step Solutions

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117 questions / 3 sections / Fully worked

Original questions, choices, tables, and graphs are followed by clear numbered steps. Key equations appear on their own lines; each solution ends with a highlighted answer. Exact fractions and radicals are retained whenever possible.

Question numbering follows the source within each section. Labels such as **2.2 / Q16** keep every solution easy to locate. A complete solution stays together; longer original questions may occupy the preceding page.

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Source: the supplied Unit 2 question packet and answer-key screenshot. All 117 answers were compared with the supplied key after solving. See the verification notes for source wording and units requiring clarification.

2.1 Equivalent Expressions

2.1 / Question 1

The equation $(ax + 3)(5x^2 - bx + 4) = 20x^3 - 9x^2 - 2x + 12$ is true for all x , where a and b are constants. What is the value of ab ?

A. 18

B. 20

C. 24

D. 40

Worked solution

2.1 / Q1

Step 1. Match the cubic coefficient.

Only $(ax)(5x^2)$ produces an x^3 term.

$$5a = 20 \quad \Rightarrow \quad a = 4.$$

Step 2. Match the quadratic coefficient.

The x^2 terms are $-abx^2$ and $15x^2$.

$$15 - ab = -9 \quad \Rightarrow \quad ab = 24.$$

Step 3. Check the remaining coefficient.

Since $a = 4$, $b = 6$. The coefficient of x is $4a - 3b = 16 - 18 = -2$, as required.

Final answer: C; 24

2.1 / Question 2

Which of the following expressions is(are) a factor of $3x^2 + 20x - 63$?

I. $x - 9$

II. $3x - 7$

A. I only

B. II only

C. I and II

D. Neither I nor II

Worked solution

2.1 / Q2

Step 1. Split the middle term.

Find two numbers with product $3(-63) = -189$ and sum 20: 27 and -7 .

$$3x^2 + 20x - 63 = 3x^2 + 27x - 7x - 63.$$

Step 2. Factor by grouping.

$$3x(x + 9) - 7(x + 9) = (3x - 7)(x + 9).$$

Step 3. Evaluate the statements.

II is a factor. I is not: $x - 9$ would require $x = 9$ to be a zero, but the polynomial equals 360 there.

Final answer: B; II only

2.1 / Question 3

If $\frac{\sqrt{x^5}}{\sqrt[3]{x^4}} = x^{\frac{a}{b}}$ for all positive values of x , what is the value of $\frac{a}{b}$?

Worked solution

2.1 / Q3

Step 1. Convert each radical to a power.

For $x > 0$,

$$\sqrt{x^5} = x^{5/2}, \quad \sqrt[3]{x^4} = x^{4/3}.$$

Step 2. Subtract exponents when dividing.

$$\frac{x^{5/2}}{x^{4/3}} = x^{5/2-4/3} = x^{15/6-8/6} = x^{7/6}.$$

Step 3. Read the requested ratio.

The exponent is a/b , so $a/b = 7/6$. Positivity of x makes these power rules valid.

Final answer: $\frac{7}{6}$

2.1 / Question 4

The equation $\frac{2}{x-2} + \frac{3}{x+5} = \frac{rx+t}{(x-2)(x+5)}$ is true for all $x > 2$, where r and t are positive constants. What is the value of rt ?

A. -20

B. 15

C. 20

D. 60

Worked solution

2.1 / Q4

Step 1. Use a common denominator.

$$\frac{2}{x-2} + \frac{3}{x+5} = \frac{2(x+5) + 3(x-2)}{(x-2)(x+5)}.$$

Step 2. Simplify the numerator.

$$2x + 10 + 3x - 6 = 5x + 4.$$

Thus $r = 5$ and $t = 4$.

Step 3. Multiply the constants.

$rt = 5(4) = 20$. The stated domain $x > 2$ excludes both denominator zeros.

Final answer: C; 20

2.1 / Question 5

For what value of x is the given expression $\sqrt[5]{70n} \left(\sqrt[6]{70n} \right)^2$ equivalent to $(70n)^{30x}$, where $n > 1$?

Worked solution

2.1 / Q5

Step 1. Convert the radicals to fractional exponents.

$$\sqrt[5]{70n} \left(\sqrt[6]{70n} \right)^2 = (70n)^{1/5} (70n)^{2/6}.$$

Step 2. Add the exponents.

$$\frac{1}{5} + \frac{2}{6} = \frac{1}{5} + \frac{1}{3} = \frac{8}{15}.$$

Step 3. Match and solve the exponents.

Since $70n > 1$, equal powers require equal exponents.

$$30x = \frac{8}{15} \quad \Rightarrow \quad x = \frac{4}{225}.$$

Final answer: $\frac{4}{225}$

2.1 / Question 6

The expression $(3x - 23)(19x + 6)$ is equivalent to the expression $ax^2 + bx + c$, where a, b, c are constants. What is the value of b ?

Worked solution

2.1 / Q6

Step 1. Expand the product.

$$(3x - 23)(19x + 6) = 57x^2 + 18x - 437x - 138.$$

Step 2. Combine the linear terms.

$$18x - 437x = -419x.$$

Step 3. Identify b .

The standard form is $57x^2 - 419x - 138$, so the coefficient b is -419 .

Final answer: -419

2.1 / Question 7

Which expression is equivalent to $\frac{4}{4x-5} - \frac{1}{x+1}$?

A. $\frac{1}{(x+1)(4x-5)}$

B. $\frac{3}{3x-6}$

C. $-\frac{1}{(x+1)(4x-5)}$

D. $\frac{9}{(x+1)(4x-5)}$

Worked solution

2.1 / Q7

Step 1. State the domain and common denominator.

The original expression requires $x \neq 5/4, -1$. Use $(4x-5)(x+1)$ as the common denominator.

Step 2. Combine the numerators.

$$\frac{4}{4x-5} - \frac{1}{x+1} = \frac{4(x+1) - (4x-5)}{(4x-5)(x+1)}.$$

Step 3. Simplify carefully.

$4x + 4 - 4x + 5 = 9$, so the result is $\frac{9}{(x+1)(4x-5)}$, on the original domain.

Final answer: D; $\frac{9}{(x+1)(4x-5)}$

2.1 / Question 8

In the expression $\frac{x^2 - c}{x - b}$, b and c are positive integers. If the expression is equivalent to $x + b$ and $x \neq b$, which of the following could be the value of c ?

A. 4

B. 6

C. 8

D. 10

Worked solution

2.1 / Q8

Step 1. Multiply by the nonzero denominator.

For $x \neq b$, equivalence gives

$$x^2 - c = (x - b)(x + b).$$

Step 2. Use the difference of squares.

The right side is $x^2 - b^2$, so $c = b^2$.

Step 3. Use the integer condition.

Because b is a positive integer, c must be a positive perfect square. Of 4, 6, 8, and 10, only 4 qualifies ($b = 2$).

Final answer: A; 4

2.1 / Question 9

The expression $0.36x^2 + 0.63x + 1.17$ can be rewritten as $a(4x^2 + 7x + 13)$, where a is a constant. What is the value of a ?

Worked solution

2.1 / Q9

Step 1. Compare the quadratic coefficients.

$$4a = 0.36 \quad \Rightarrow \quad a = 0.09.$$

Step 2. Check the linear coefficient.

$$7(0.09) = 0.63.$$

Step 3. Check the constant term.

$$13(0.09) = 1.17. \text{ All three coefficients agree, so } a = 0.09 = 9/100.$$

Final answer: $0.09 = \frac{9}{100}$

2.1 / Question 10

In the expression $3(2x^2 + px + 8) - 16x(p + 4)$, p is a constant. This expression is equivalent to $6x^2 - 155x + 24$. What is the value of p ?

A. -3

B. 7

C. 13

D. 155

Worked solution

2.1 / Q10

Step 1. Expand the expression.

$$3(2x^2 + px + 8) - 16x(p + 4) = 6x^2 + (3p - 16p - 64)x + 24.$$

Step 2. Match the coefficient of x .

$$-13p - 64 = -155 \quad \Rightarrow \quad -13p = -91.$$

Step 3. Solve and verify.

$p = 7$. Then $-13(7) - 64 = -155$, and the other coefficients already match.

Final answer: B; 7

2.1 / Question 11

The expression $\frac{x^{-2}y^{\frac{1}{2}}}{x^{\frac{1}{3}}y^{-1}}$, where $x > 1$ and $y > 1$, is equivalent to which of the following?

A. $\frac{\sqrt{y}}{\sqrt[3]{x^2}}$

B. $\frac{y\sqrt{y}}{\sqrt[3]{x^2}}$

C. $\frac{y\sqrt{y}}{x\sqrt{x}}$

D. $\frac{y\sqrt{y}}{x^2\sqrt[3]{x}}$

Worked solution

2.1 / Q11

Step 1. Subtract exponents for each variable.

$$\frac{x^{-2}y^{1/2}}{x^{1/3}y^{-1}} = x^{-2-1/3}y^{1/2-(-1)} = x^{-7/3}y^{3/2}.$$

Step 2. Rewrite with positive exponents.

$$x^{-7/3}y^{3/2} = \frac{y^{3/2}}{x^{7/3}}.$$

Step 3. Convert to radicals.

Since $x, y > 1$,

$$\frac{y^{3/2}}{x^{7/3}} = \frac{y\sqrt{y}}{x^2\sqrt[3]{x}}.$$

Final answer: D; $\frac{y\sqrt{y}}{x^2\sqrt[3]{x}}$

2.1 / Question 12

The expression $4x^2 + bx - 45$, where b is a constant, can be rewritten as $(hx + k)(x + j)$, where h, k , and j are integers. Which of the following must be an integer?

A. $\frac{b}{h}$

B. $\frac{b}{k}$

C. $\frac{45}{h}$

D. $\frac{45}{k}$

Worked solution

2.1 / Q12

Step 1. Expand the factored form.

$$(hx + k)(x + j) = hx^2 + (hj + k)x + kj.$$

Step 2. Match coefficients. $h = 4$, $b = 4j + k$, and $kj = -45$. In particular, $k \neq 0$.**Step 3. Identify the guaranteed integer.**

$$\frac{45}{k} = -j,$$

which is an integer. For comparison, $k = 5, j = -9$ gives $b = -31$, so b/h and b/k need not be integers; $45/h = 45/4$ is not an integer.

Final answer: D; $\frac{45}{k}$

2.1 / Question 13

Let $f(x) = x^3 - 9x$, $g(x) = x^2 - 2x - 3$. Which of the following expressions is $\frac{f(x)}{g(x)}$ equivalent to, for $x > 3$?

A. $\frac{1}{x+1}$

B. $\frac{x+3}{x+1}$

C. $\frac{x(x-3)}{x+1}$

D. $\frac{x(x+3)}{x+1}$

Worked solution

2.1 / Q13

Step 1. Factor numerator and denominator.

$$x^3 - 9x = x(x-3)(x+3), \quad x^2 - 2x - 3 = (x-3)(x+1).$$

Step 2. Cancel the common factor.

Since $x > 3$, the factor $x - 3$ is nonzero.

$$\frac{f(x)}{g(x)} = \frac{x(x-3)(x+3)}{(x-3)(x+1)} = \frac{x(x+3)}{x+1}.$$

Step 3. Check the domain.

For $x > 3$, $x + 1$ is also nonzero. The simplification is valid throughout the stated domain.

Final answer: D; $\frac{x(x+3)}{x+1}$

2.1 / Question 14

The expression $\frac{1}{3}x^2 - 2$ can be rewritten as $\frac{1}{3}(x - k)(x + k)$, where k is a positive constant. What is the value of k ?

A. 2

B. 6

C. $\sqrt{2}$

D. $\sqrt{6}$

Worked solution

2.1 / Q14

Step 1. Factor out $1/3$.

$$\frac{1}{3}x^2 - 2 = \frac{1}{3}(x^2 - 6).$$

Step 2. Compare with the proposed factors.

$$\frac{1}{3}(x - k)(x + k) = \frac{1}{3}(x^2 - k^2).$$

Thus $k^2 = 6$.

Step 3. Choose the positive value.

The problem specifies $k > 0$, so $k = \sqrt{6}$.

Final answer: D; $\sqrt{6}$

2.1 / Question 15

Which of the following is equivalent to $\left(a + \frac{b}{2}\right)^2$?

A. $a^2 + \frac{b^2}{2}$

B. $\frac{a^2 + b^2}{4}$

C. $\frac{a^2 + ab}{2} + \frac{b^2}{2}$

D. $a^2 + ab + \frac{b^2}{4}$

Worked solution

2.1 / Q15

Step 1. Use the square-of-a-sum identity.

$(u + v)^2 = u^2 + 2uv + v^2$. Here $u = a$ and $v = b/2$.

Step 2. Expand all three terms.

$$\left(a + \frac{b}{2}\right)^2 = a^2 + 2a\left(\frac{b}{2}\right) + \left(\frac{b}{2}\right)^2.$$

Step 3. Simplify.

$$a^2 + ab + \frac{b^2}{4}.$$

The middle term ab must be included.

Final answer: D; $a^2 + ab + \frac{b^2}{4}$

2.1 / Question 16

Which expression is equivalent to $\frac{y+12}{x-8} + \frac{y(x-8)}{x^2y-8xy}$?

A. $\frac{xy+y+4}{x^2y-16x+64xy}$

B. $\frac{xy+9+12}{x^2y-8xy+x-8}$

C. $\frac{xy^2+13xy-8y}{x^2y-8xy}$

D. $\frac{xy^2+13xy-8y}{x^2y-16x^2y+64xy}$

Worked solution

2.1 / Q16

Step 1. Factor the denominator and preserve restrictions.

$x^2y - 8xy = xy(x - 8)$, so the original domain requires $x \neq 0, 8$ and $y \neq 0$.

Step 2. Use the common denominator.

$$\frac{y+12}{x-8} + \frac{y(x-8)}{xy(x-8)} = \frac{xy(y+12) + y(x-8)}{xy(x-8)}.$$

Step 3. Expand and combine the numerator.

$$xy^2 + 12xy + xy - 8y = xy^2 + 13xy - 8y.$$

This gives choice C, with the original domain retained.

Final answer: C; $\frac{xy^2+13xy-8y}{x^2y-8xy}$

2.1 / Question 17

One of the factors of $2x^3 + 42x^2 + 208x$ is $x + b$, where b is a positive constant. What is the smallest possible value of b ?

Worked solution

2.1 / Q17

Step 1. Factor out the greatest common factor.

$$2x^3 + 42x^2 + 208x = 2x(x^2 + 21x + 104).$$

Step 2. Factor the quadratic.

Since $8 + 13 = 21$ and $8(13) = 104$,

$$2x(x^2 + 21x + 104) = 2x(x + 8)(x + 13).$$

Step 3. Choose the smallest positive b .

The factors of the form $x + b$ with $b > 0$ correspond to $b = 8$ and $b = 13$. The smaller value is 8.

Final answer: 8

2.1 / Question 18

The equation $\frac{x^2 + 6x - 7}{x + 7} = ax + d$ is true for all $x \neq -7$, where a and d are integers. What is the value of $a + d$?

A. -6

B. -1

C. 0

D. 1

Worked solution

2.1 / Q18

Step 1. Factor the numerator.

$$x^2 + 6x - 7 = (x + 7)(x - 1).$$

Step 2. Cancel on the stated domain.

For $x \neq -7$,

$$\frac{(x + 7)(x - 1)}{x + 7} = x - 1.$$

Step 3. Match coefficients.

$a = 1$ and $d = -1$, so $a + d = 0$.

Final answer: C; 0

2.1 / Question 19

Which of the following expressions is equivalent to $\frac{x^2 - 2x - 5}{x - 3}$?

A. $x - 5 - \frac{20}{x - 3}$

B. $x - 5 - \frac{10}{x - 3}$

C. $x + 1 - \frac{8}{x - 3}$

D. $x + 1 - \frac{2}{x - 3}$

Worked solution

2.1 / Q19

Step 1. Divide the leading terms.

Dividing x^2 by x gives x . Subtract $x(x - 3) = x^2 - 3x$ from the numerator to obtain $x - 5$.

Step 2. Complete the division.

The next quotient term is 1. Subtract $(x - 3)$ from $(x - 5)$ to obtain the remainder -2 .

Step 3. Write and verify the result.

$$\frac{x^2 - 2x - 5}{x - 3} = x + 1 - \frac{2}{x - 3}, \quad x \neq 3.$$

Check: $(x - 3)(x + 1) - 2 = x^2 - 2x - 5$.

Final answer: D; $x + 1 - \frac{2}{x - 3}$

2.1 / Question 20

The expression $(7532 + 100y^2) + 10(10y^2 - 110)$ can be written in the form $ay^2 + b$, where a and b are constants. What is the value of $a + b$?

Worked solution

2.1 / Q20

Step 1. Distribute the factor 10.

$$(7532 + 100y^2) + 10(10y^2 - 110) = 7532 + 100y^2 + 100y^2 - 1100.$$

Step 2. Combine like terms.

The expression becomes $200y^2 + 6432$. Thus $a = 200$ and $b = 6432$.

Step 3. Add the coefficients requested.

$$a + b = 200 + 6432 = 6632.$$

Final answer: 6632

2.1 / Question 21

The expression $6\sqrt[5]{3^5x^{45}} \cdot \sqrt[8]{2^8x}$ is equivalent to ax^b , where a and b are positive constants and $x > 1$. What is the value of $a + b$?

Worked solution

2.1 / Q21

Step 1. Simplify the fifth root.

Since $x > 1$,

$$6\sqrt[5]{3^5x^{45}} = 6(3x^9) = 18x^9.$$

Step 2. Simplify the eighth root and multiply.

$$\sqrt[8]{2^8x} = 2x^{1/8}, \quad (18x^9)(2x^{1/8}) = 36x^{73/8}.$$

Step 3. Identify and add a and b .

$$a = 36, \quad b = \frac{73}{8}, \quad a + b = \frac{288 + 73}{8} = \frac{361}{8} = 45.125.$$

Final answer: $\frac{361}{8} = 45.125$

2.1 / Question 22

If $a = c + d$, which of the following is equivalent to the expression $x^2 - c^2 - 2cd - d^2$?

A. $(x + a)^2$

B. $(x - a)^2$

C. $(x + a)(x - a)$

D. $x^2 - ax - a^2$

Worked solution

2.1 / Q22

Step 1. Recognize a perfect square.

$$c^2 + 2cd + d^2 = (c + d)^2 = a^2.$$

Step 2. Rewrite the expression.

$$x^2 - c^2 - 2cd - d^2 = x^2 - a^2.$$

Step 3. Factor the difference of squares.

$$x^2 - a^2 = (x + a)(x - a).$$

Final answer: C; $(x + a)(x - a)$

2.1 / Question 23

If a and c are positive numbers, which of the following is equivalent to $\sqrt{(a+c)^3} \cdot \sqrt{a+c}$?

A. $a + c$

B. $a^2 + c^2$

C. $a^2 + 2ac + c^2$

D. a^2c^2

Worked solution

2.1 / Q23

Step 1. Combine the radicals.

Since $a + c > 0$,

$$\sqrt{(a+c)^3} \sqrt{a+c} = \sqrt{(a+c)^4}.$$

Step 2. Simplify the square root.

$\sqrt{(a+c)^4} = (a+c)^2$, since a square is nonnegative.

Step 3. Expand the binomial.

$$(a+c)^2 = a^2 + 2ac + c^2.$$

Final answer: C; $a^2 + 2ac + c^2$

2.2 Nonlinear Equations and Systems

2.2 / Question 1

What is the positive solution to the equation $2|4 - x| + 3|4 - x| = 25$?

Worked solution

2.2 / Q1

Step 1. Combine identical absolute-value terms.

$$2|4 - x| + 3|4 - x| = 25 \quad \Rightarrow \quad |4 - x| = 5.$$

Step 2. Solve both branches.

$4 - x = 5$ gives $x = -1$; $4 - x = -5$ gives $x = 9$.

Step 3. Select and check the positive root.

The positive solution is 9. Substitution gives $2(5) + 3(5) = 25$.

Final answer: 9

2.2 / Question 2

One solution to the equation $x^2 - 2x - 9 = 0$ can be written as $1 + \sqrt{k}$, where k is a constant. What is the value of k ?

A. 8

B. 10

C. 20

D. 40

Worked solution

2.2 / Q2

Step 1. Complete the square.

$$x^2 - 2x - 9 = 0 \quad \Rightarrow \quad (x - 1)^2 = 10.$$

Step 2. Take both square roots.

$$x = 1 \pm \sqrt{10}.$$

Step 3. Match the specified form.

The solution $1 + \sqrt{k}$ has $k = 10$.

Final answer: B; 10

2.2 / Question 3

In the xy -plane, a line with equation $2y = 4.5$ intersects a parabola at exactly one point. If the parabola has equation $y = -4x^2 + bx$, where b is a positive constant, what is the value of b ?

Worked solution

2.2 / Q3

Step 1. Set the parabola equal to the line.

The line is $y = 4.5/2 = 9/4$. Thus

$$4x^2 - bx + \frac{9}{4} = 0.$$

Step 2. Require a repeated root.

Exactly one intersection requires discriminant zero.

$$(-b)^2 - 4(4)\left(\frac{9}{4}\right) = b^2 - 36 = 0.$$

Step 3. Use positivity and check.

$b = 6$. Then $y = -4(x - 3/4)^2 + 9/4$, so the horizontal line meets exactly the vertex.

Final answer: 6

2.2 / Question 4

Given the system:

$$\begin{cases} x - y = 1 \\ x + y = x^2 - 3 \end{cases}$$

Which ordered pair is a solution?

A. $(1 + \sqrt{3}, \sqrt{3})$

B. $(\sqrt{3}, -\sqrt{3})$

C. $(1 + \sqrt{5}, \sqrt{5})$

D. $(\sqrt{5}, -1 + \sqrt{5})$

Worked solution

2.2 / Q4

Step 1. Express y in terms of x .

From $x - y = 1$, obtain $y = x - 1$.

Step 2. Substitute and solve.

$$x + (x - 1) = x^2 - 3 \Rightarrow x^2 - 2x - 2 = 0.$$

Hence $x = 1 \pm \sqrt{3}$ and $y = \pm\sqrt{3}$ with matching signs.

Step 3. Identify the listed pair.

Choice A is $(1 + \sqrt{3}, \sqrt{3})$. It satisfies both $x - y = 1$ and $x + y = x^2 - 3$.

Final answer: A; $(1 + \sqrt{3}, \sqrt{3})$

2.2 / Question 5

If x is a solution to the equation $\frac{1}{x^2 + 10x + 25} = 4$, which of the following is a possible value of $x + 5$?

A. $\frac{1}{2}$

B. $\frac{5}{2}$

C. $\frac{9}{2}$

D. $\frac{11}{2}$

Worked solution

2.2 / Q5

Step 1. Recognize the denominator.

$$x^2 + 10x + 25 = (x + 5)^2, \quad x \neq -5.$$

Step 2. Solve for the squared expression.

$$\frac{1}{(x + 5)^2} = 4 \quad \Rightarrow \quad (x + 5)^2 = \frac{1}{4}.$$

Step 3. Take both signs and select a listed value.

$x + 5 = \pm 1/2$. Only $1/2$ is listed, and its square makes the reciprocal equal 4.

Final answer: A; $\frac{1}{2}$

2.2 / Question 6

During a 5-second time interval, the average acceleration a (m/s²) of an object with an initial velocity of 12 m/s is defined by the equation

$$a = \frac{v_f - 12}{5}$$

If this is rewritten as $v_f = xa + y$, what is the value of x ?

Worked solution

2.2 / Q6

Step 1. Clear the denominator.

$$a = \frac{v_f - 12}{5} \quad \Rightarrow \quad 5a = v_f - 12.$$

Step 2. Isolate final velocity.

$$v_f = 5a + 12.$$

Step 3. Match coefficients.

Comparing with $v_f = xa + y$ gives $x = 5$ and $y = 12$. The requested value is 5.

Final answer: 5

2.2 / Question 7

Which of the following is a solution to $2x^2 - 2 = 2x + 3$?

A. 2

B. $1 - \sqrt{11}$

C. $\frac{1}{2} + \sqrt{11}$

D. $\frac{1 + \sqrt{11}}{2}$

Worked solution

2.2 / Q7

Step 1. Put the equation in standard form.

$$2x^2 - 2 = 2x + 3 \quad \Rightarrow \quad 2x^2 - 2x - 5 = 0.$$

Step 2. Apply the quadratic formula.

$$x = \frac{2 \pm \sqrt{(-2)^2 - 4(2)(-5)}}{4} = \frac{2 \pm \sqrt{44}}{4}.$$

Step 3. Simplify the radical.

$$x = \frac{1 \pm \sqrt{11}}{2}.$$

The positive-sign root is listed in D.

Final answer: D; $\frac{1+\sqrt{11}}{2}$

2.2 / Question 8

In the xy -plane, a line with equation $2y = c$ intersects a parabola $y = -2x^2 + 9x$ at exactly one point. What is the value of c ?

Worked solution

2.2 / Q8

Step 1. Find the vertex's horizontal coordinate.

For $y = -2x^2 + 9x$,

$$x_v = -\frac{9}{2(-2)} = \frac{9}{4}.$$

Step 2. Evaluate the maximum height.

$$y_v = -2\left(\frac{9}{4}\right)^2 + 9\left(\frac{9}{4}\right) = \frac{81}{8}.$$

Step 3. Relate the line to the vertex.

The horizontal line is $y = c/2$. One intersection requires $c/2 = 81/8$, so $c = 81/4 = 20.25$.

Final answer: $\frac{81}{4} = 20.25$

2.2 / Question 9

What is the smallest solution to the equation

$$\sqrt{(x-2)^2} = \sqrt{3x+34}$$

Worked solution

2.2 / Q9

Step 1. State the domain and square the equality.

The right radical requires $x \geq -34/3$. Squaring the nonnegative sides gives

$$(x-2)^2 = 3x+34.$$

Step 2. Solve the resulting quadratic.

$$x^2 - 7x - 30 = 0 \quad \Rightarrow \quad (x-10)(x+3) = 0.$$

Candidates are 10 and -3 .

Step 3. Check both in the original equation.

At $x = -3$, both sides equal 5; at $x = 10$, both equal 8. Both are valid, and the smaller is -3 .

Final answer: -3

2.2 / Question 10

Given the equation

$$57x^2 + (57b + a)x + ab = 0$$

where a and b are positive constants. The product of the solutions is kab . What is the value of k ?

A. $\frac{1}{57}$

B. $\frac{1}{19}$

C. 1

D. 57

Worked solution

2.2 / Q10

Step 1. Identify the quadratic coefficients.

For $Ax^2 + Bx + C = 0$, the product of its roots is C/A . Here $A = 57$ and $C = ab$.

Step 2. Compute the product.

$$x_1x_2 = \frac{ab}{57} = kab.$$

Step 3. Solve for k .

Since $a, b > 0$, divide by ab to get $k = 1/57$. Equivalently, the polynomial factors as $(57x + a)(x + b)$.

Final answer: A; $\frac{1}{57}$

2.2 / Question 11

In the equation $-x^2 + bx - 676 = 0$, b is a positive integer and the equation has no real solution. What is the greatest possible value of b ?

Worked solution

2.2 / Q11

Step 1. Compute the discriminant.

For $-x^2 + bx - 676 = 0$,

$$\Delta = b^2 - 4(-1)(-676) = b^2 - 2704.$$

Step 2. Require no real solutions.

$\Delta < 0$ gives $b^2 < 2704 = 52^2$. Since b is positive, $b < 52$.

Step 3. Choose the greatest integer.

The greatest possible integer is 51. Check: $51^2 - 2704 = -103 < 0$; at 52 the discriminant is zero.

Final answer: 51

2.2 / Question 12

If $3x^2 - 18x - 15 = 0$, what is the value of $x^2 - 6x$?

Worked solution

2.2 / Q12

Step 1. Group the requested expression.

$$3x^2 - 18x - 15 = 0 \quad \Rightarrow \quad 3(x^2 - 6x) = 15.$$

Step 2. Divide by 3.

$$x^2 - 6x = 5.$$

Step 3. Check why individual roots are unnecessary.

Every solution of the original equation satisfies this rearrangement, so the requested expression has value 5 for either root.

Final answer: 5

2.2 / Question 13

In the equation $2x^2 - 4x = t$, t is a constant. If the equation has no real solutions, which of the following could be the value of t ?

A. -3

B. -1

C. 1

D. 3

Worked solution

2.2 / Q13

Step 1. Complete the square.

$$2x^2 - 4x = 2(x - 1)^2 - 2.$$

Step 2. Identify the range.

Because $(x - 1)^2 \geq 0$, the left side is at least -2 . There are no real solutions when $t < -2$.

Step 3. Compare the choices.

Only $t = -3$ is below -2 . At that value, the equation would require a negative square.

Final answer: A; -3

2.2 / Question 14

In the equation $x(kx - 56) = -16$, k is an integer constant. If the equation has no real solution, what is the least possible value of k ?

Worked solution

2.2 / Q14

Step 1. Write the quadratic and handle $k = 0$.

$$kx^2 - 56x + 16 = 0.$$

If $k = 0$, the equation is linear and has a solution, so that case is excluded.

Step 2. Apply the discriminant condition.

For $k \neq 0$, no real solution requires

$$(-56)^2 - 4k(16) < 0 \quad \Rightarrow \quad 3136 - 64k < 0.$$

Step 3. Choose the smallest integer.

$k > 49$, so the smallest integer is 50. At $k = 49$, there is a repeated real root; at 50, the discriminant is -64 .

Final answer: 50

2.2 / Question 15

The equation $\frac{14x}{7y} = 2\sqrt{w+19}$ relates the distinct positive real numbers w, x , and y . Which equation correctly expresses w in terms of x and y ?

A. $\sqrt{\frac{x}{y}} - 19$

B. $\sqrt{\frac{28x}{14y}} - 19$

C. $\left(\frac{x}{y}\right)^2 - 19$

D. $\left(\frac{28x}{14y}\right)^2 - 19$

Worked solution

2.2 / Q15

Step 1. Simplify and isolate the radical.

The original radical covers $w + 19$.

$$\frac{14x}{7y} = 2\sqrt{w+19} \quad \Rightarrow \quad \frac{x}{y} = \sqrt{w+19}.$$

Step 2. Square both sides.

Since $x, y > 0$, both sides are nonnegative.

$$\left(\frac{x}{y}\right)^2 = w + 19.$$

Step 3. Solve for w .

$$w = \left(\frac{x}{y}\right)^2 - 19.$$

The given positivity of w further implies $(x/y)^2 > 19$.

Final answer: C; $w = \left(\frac{x}{y}\right)^2 - 19$

2.2 / Question 16

Solve for x : $\sqrt{2x+6} + 4 = x + 3$ What is the solution set of the equation above?

A. $\{-1\}$

B. $\{5\}$

C. $\{-1, 5\}$

D. $\{0, -1, 5\}$

Worked solution

2.2 / Q16

Step 1. Isolate the radical and restrict x .

$$\sqrt{2x+6} = x - 1.$$

Since a square root is nonnegative, any solution must have $x \geq 1$.

Step 2. Square and factor.

$$2x + 6 = (x - 1)^2 \quad \Rightarrow \quad x^2 - 4x - 5 = (x - 5)(x + 1) = 0.$$

Step 3. Reject the extraneous candidate.

$x = -1$ violates $x \geq 1$. For $x = 5$, the original sides are $\sqrt{16} + 4 = 8$ and $5 + 3 = 8$. Thus the solution set is $\{5\}$.

Final answer: B; $\{5\}$

2.2 / Question 17

The function f , defined by

$$f(x) = -\frac{1}{2}(x - 4)^2 + 10$$

If the function g (not shown) is defined by $g(x) = -x + 10$, what is one possible value of a such that $f(a) = g(a)$?

Worked solution

2.2 / Q17

Step 1. Set the functions equal.

$$-\frac{1}{2}(a - 4)^2 + 10 = -a + 10 \quad \Rightarrow \quad (a - 4)^2 = 2a.$$

Step 2. Expand and factor.

$$a^2 - 10a + 16 = 0 \quad \Rightarrow \quad (a - 2)(a - 8) = 0.$$

Step 3. Check and give one possible value.

At $a = 2$, both functions equal 8; at $a = 8$, both equal 2. Either 2 or 8 answers the question.

Final answer: 2 or 8

2.2 / Question 18

In the equation $x^2 - 34x + c = 0$, c is a constant. The equation has no real solutions if $c > n$. What is the least possible value of n ?

Worked solution

2.2 / Q18

Step 1. Complete the square.

$$x^2 - 34x + c = 0 \quad \Rightarrow \quad (x - 17)^2 = 289 - c.$$

Step 2. Determine when real roots are impossible.

The left side is nonnegative, so no real solutions occur exactly when $289 - c < 0$, or $c > 289$.

Step 3. Identify the least threshold.

$n = 289$. Any smaller threshold would incorrectly include some values of c at or below 289, where real solutions exist.

Final answer: 289

2.2 / Question 19

In the equation $-16x^2 - 8x + c = 0$, c is a constant. The equation has exactly one solution. What is the value of c ?

Worked solution

2.2 / Q19

Step 1. Complete the square.

$$-16x^2 - 8x + c = -16\left(x + \frac{1}{4}\right)^2 + (c + 1).$$

Step 2. Require exactly one real root.

The equation gives $(x + 1/4)^2 = (c + 1)/16$. A single root occurs only when the right side is zero.

Step 3. Solve and check.

$c = -1$. The equation becomes $-16(x + 1/4)^2 = 0$, with the unique solution $x = -1/4$.

Final answer: -1

2.2 / Question 20

How many distinct real solutions does the equation $(x - 1)^2 = -4$ have?

A. Exactly one

B. Exactly two

C. Infinitely many

D. Zero

Worked solution

2.2 / Q20

Step 1. Use the sign of a real square.

For every real x , $(x - 1)^2 \geq 0$.

Step 2. Compare with the required value.

The equation requires $(x - 1)^2 = -4$, a negative value.

Step 3. Count the real solutions.

No real number can satisfy the equation. The number of distinct real solutions is zero.

Final answer: D; zero real solutions

2.2 / Question 21

How many distinct real solutions does the equation $5x^2 + 10x + 16 = 0$ have?

A. Exactly one

B. Exactly two

C. Infinitely many

D. Zero

Worked solution

2.2 / Q21

Step 1. Compute the discriminant.

$$\Delta = 10^2 - 4(5)(16) = 100 - 320 = -220.$$

Step 2. Apply the real-root criterion.

A negative discriminant means there are no real roots.

Step 3. Confirm by completing the square.

$5x^2 + 10x + 16 = 5(x + 1)^2 + 11 > 0$ for all real x , so it can never equal zero.

Final answer: D; zero real solutions

2.2 / Question 22

Given the system:

$$\begin{cases} y = x^2 + 2x + 1 \\ x + y + 1 = 0 \end{cases}$$

If (x_1, y_1) and (x_2, y_2) are the two solutions to the system, what is the value of $y_1 + y_2$?

A. -3

B. -2

C. -1

D. 1

Worked solution

2.2 / Q22

Step 1. Substitute the linear equation.

From $x + y + 1 = 0$, $y = -x - 1$. Thus

$$x^2 + 2x + 1 = -x - 1.$$

Step 2. Find both intersections.

$x^2 + 3x + 2 = (x + 1)(x + 2) = 0$, giving $(x, y) = (-1, 0)$ and $(-2, 1)$.

Step 3. Add the y -coordinates.

$$y_1 + y_2 = 0 + 1 = 1.$$

Final answer: D; 1

2.2 / Question 23

If $u - 3 = \frac{6}{t - 2}$, what is t in terms of u ?

A. $\frac{1}{u}$

B. $\frac{2u + 9}{u}$

C. $\frac{1}{u - 3}$

D. $\frac{2u}{u - 3}$

Worked solution

2.2 / Q23

Step 1. Clear the denominator.

The original equation requires $t \neq 2$.

$$(u - 3)(t - 2) = 6.$$

It also follows that $u \neq 3$.

Step 2. Isolate t .

$$t = 2 + \frac{6}{u - 3}.$$

Step 3. Combine into one fraction.

$$t = \frac{2(u - 3) + 6}{u - 3} = \frac{2u}{u - 3}.$$

For $u \neq 3$, this expression cannot equal 2, so the original restriction is respected.

Final answer: D; $t = \frac{2u}{u-3}$

2.2 / Question 24

In the equation $-9x^2 + 30x + c = 0$, c is a constant. The equation has exactly one solution. What is the value of c ?

A. 3

B. 0

C. -25

D. -53

Worked solution

2.2 / Q24

Step 1. Compute the discriminant.

$$\Delta = 30^2 - 4(-9)c = 900 + 36c.$$

Step 2. Set it equal to zero.

Exactly one real solution requires $900 + 36c = 0$.

Step 3. Solve and verify.

$c = -25$. The resulting equation is $-(3x - 5)^2 = 0$, with one root $x = 5/3$.

Final answer: C; -25

2.2 / Question 25

In the equation

$$\frac{x^2}{\sqrt{x^2 - c^2}} = \frac{c^2}{\sqrt{x^2 - c^2}} + 39,$$

where c is a positive constant, which of the following is one of the solutions?

A. $-c$

B. $-c^2 - 39^2$

C. $-\sqrt{39^2 - c^2}$

D. $-\sqrt{c^2 + 39^2}$

Worked solution

2.2 / Q25

Step 1. State the domain and combine fractions.

The radical is in a denominator, so $x^2 - c^2 > 0$. Subtract the two fractions:

$$\frac{x^2 - c^2}{\sqrt{x^2 - c^2}} = 39.$$

Step 2. Simplify and solve.

$$\sqrt{x^2 - c^2} = 39 \quad \Rightarrow \quad x^2 = c^2 + 39^2.$$

Step 3. Select a valid listed root.

Both signs are valid: $x = \pm\sqrt{c^2 + 39^2}$. Each gives a nonzero denominator of 39. The negative root is choice D.

Final answer: D; $-\sqrt{c^2 + 39^2}$

2.2 / Question 26

The graphs of the equations

$$\begin{cases} 8x + y = -11 \\ 2x^2 = y + 341 \end{cases}$$

intersect at the point (x, y) in the xy -plane. What is a possible value of x ?

A. -15

B. -11

C. 2

D. 8

Worked solution

2.2 / Q26

Step 1. Express y from each equation.

$$y = -11 - 8x, \quad y = 2x^2 - 341.$$

Step 2. Equate and factor.

$$2x^2 + 8x - 330 = 0 \quad \Rightarrow \quad (x + 15)(x - 11) = 0.$$

Step 3. Select and check the listed value.

The candidates are -15 and 11 ; only -15 is listed. It gives $y = 109$, and $2(-15)^2 = 109 + 341 = 450$.

Final answer: A; -15

2.2 / Question 27

What is the sum of the solutions to the equation

$$x(x + 1) - 56 = 4x(x - 7)?$$

Worked solution

2.2 / Q27

Step 1. Expand both sides.

$$x^2 + x - 56 = 4x^2 - 28x.$$

Step 2. Write standard form.

$$3x^2 - 29x + 56 = 0.$$

Step 3. Use the sum-of-roots formula.

The sum is $-B/A = 29/3$. Its discriminant is $29^2 - 4(3)(56) = 169 > 0$, confirming two real roots.

Final answer: $\frac{29}{3}$

2.2 / Question 28

In the xy -plane, the graph of $y = x^2 - 9$ intersects line p at $(1, a)$ and $(5, b)$, where a and b are constants. What is the slope of line p ?

A. 6

B. 2

C. -2

D. -6

Worked solution

2.2 / Q28

Step 1. Find the two points on the parabola.

At $x = 1$, $a = 1^2 - 9 = -8$. At $x = 5$, $b = 5^2 - 9 = 16$.

Step 2. Calculate the slope.

$$m = \frac{16 - (-8)}{5 - 1} = \frac{24}{4} = 6.$$

Step 3. Check the secant line.

The line $y = 6x - 14$ gives -8 at $x = 1$ and 16 at $x = 5$, matching both points.

Final answer: A; 6

2.2 / Question 29

The formula

$$D = T - \frac{9}{25}(100 - H)$$

approximates the dew point D in Fahrenheit, given temperature T in Fahrenheit, and relative humidity $H\%$, where $H > 50$. Which of the following expresses H in terms of D and T ?

A. $H = \frac{25}{9}(D - T) + 100$

B. $H = \frac{25}{9}(D - T) - 100$

C. $H = \frac{25}{9}(D + T) + 100$

D. $H = \frac{25}{9}(D + T) - 100$

Worked solution

2.2 / Q29

Step 1. Expand the formula.

$$D = T - 36 + \frac{9}{25}H.$$

Step 2. Isolate the humidity term.

$$D - T + 36 = \frac{9}{25}H.$$

Step 3. Multiply by $25/9$.

$$H = \frac{25}{9}(D - T) + 100.$$

The relation is used within the stated range $H > 50$.

Final answer: A; $H = \frac{25}{9}(D - T) + 100$

2.2 / Question 30

Given the equation

$$x - 29 = (x - a)(x - 29),$$

where a is a constant and $a > 30$, which of the following are solutions?

- I. a
- II. $a + 1$
- III. 29

A. I and II only

B. I and III only

C. II and III only

D. I, II, and III

Worked solution

2.2 / Q30

Step 1. Move all terms to one side.

$$(x - a)(x - 29) - (x - 29) = 0.$$

Step 2. Factor without dividing by $x - 29$.

$$(x - 29)(x - a - 1) = 0.$$

Thus $x = 29$ or $x = a + 1$.

Step 3. Evaluate the statements.

II and III are solutions. I is not: at $x = a$, the right side is 0 but the left is $a - 29 > 1$.

Final answer: C; II and III only

2.2 / Question 31

Given the system:

$$\begin{cases} y = 2x^2 - 21x + 64 \\ y = 3x + a \end{cases}$$

In the system of the equations, a is a constant. The graphs intersect at exactly one point (x, y) . What is the value of x ?

A. -8

B. -6

C. 6

D. 8

Worked solution

2.2 / Q31

Step 1. Set the two equations equal.

$$2x^2 - 21x + 64 = 3x + a \quad \Rightarrow \quad 2x^2 - 24x + 64 - a = 0.$$

Step 2. Complete the square.

$$2(x - 6)^2 = a + 8.$$

Step 3. Use the single-intersection condition.

A single real root requires $a = -8$, making $(x - 6)^2 = 0$. Therefore the intersection has $x = 6$.

Final answer: C; 6

2.2 / Question 32

In the xy -plane, the graph of $y = 3x^2 - 14x$ intersects the graph of $y = x$ at the points $(0, 0)$ and (a, a) . What is the value of a ?

Worked solution

2.2 / Q32

Step 1. Set the outputs equal.

$$3x^2 - 14x = x \quad \Rightarrow \quad 3x^2 - 15x = 0.$$

Step 2. Factor.

$$3x(x - 5) = 0,$$

so $x = 0$ or $x = 5$.

Step 3. Identify the second point.

The first point is already $(0, 0)$. The other is $(5, 5)$, so $a = 5$.

Final answer: 5

2.2 / Question 33

Given the equation

$$64x^2 - (16a + 4b)x + ab = 0$$

The sum of the solutions is $k(4a + b)$. What is the value of k ?

Worked solution

2.2 / Q33

Step 1. Apply the sum-of-roots formula.

$$x_1 + x_2 = \frac{16a + 4b}{64}.$$

Step 2. Factor and simplify.

$$\frac{16a + 4b}{64} = \frac{4(4a + b)}{64} = \frac{1}{16}(4a + b).$$

Step 3. Identify the coefficient.

The requested multiplier is $k = 1/16 = 0.0625$.

Final answer: $\frac{1}{16} = 0.0625$

2.2 / Question 34

What is the sum of the solutions to the equation

$$x^2 - 40x - 10 = 0$$

A. 0

B. 5

C. 10

D. 40

Worked solution

2.2 / Q34

Step 1. Identify the coefficients.

For $x^2 - 40x - 10 = 0$, $A = 1$ and $B = -40$.

Step 2. Use the sum of roots.

$$x_1 + x_2 = -\frac{B}{A} = -\frac{-40}{1} = 40.$$

Step 3. Check with the explicit roots.

Completing the square gives $x = 20 \pm \sqrt{410}$; their sum is 40.

Final answer: D; 40

2.2 / Question 35

Given the system:

$$\begin{cases} y = x^2 + 3x - 7 \\ y - 5x + 8 = 0 \end{cases}$$

How many solutions are there?

A. Exactly 4

B. Exactly 2

C. Exactly 1

D. None

Worked solution

2.2 / Q35

Step 1. Rewrite the linear equation.

$y - 5x + 8 = 0$ gives $y = 5x - 8$.

Step 2. Equate and factor.

$$x^2 + 3x - 7 = 5x - 8 \Rightarrow (x - 1)^2 = 0.$$

Step 3. Count and check the intersection.

Only $x = 1$ is possible, giving $y = -3$ in both equations. There is exactly one solution.

Final answer: C; exactly one solution

2.2 / Question 36

Solve for x :

$$\frac{4x^2}{x^2 - 9} - \frac{2x}{x + 3} = \frac{1}{x - 3}$$

A. -3

B. $-\frac{1}{2}$

C. $\frac{1}{2}$

D. 3

Worked solution

2.2 / Q36

Step 1. State restrictions and clear denominators.

The domain excludes $x = \pm 3$. Multiply by $(x - 3)(x + 3)$:

$$4x^2 - 2x(x - 3) = x + 3.$$

Step 2. Simplify and factor.

$$2x^2 + 5x - 3 = (2x - 1)(x + 3) = 0.$$

Step 3. Discard the forbidden value.

$x = -3$ makes original denominators zero. The valid root is $x = 1/2$; substitution gives $-2/5$ on both sides of the original equation.

Final answer: C; $\frac{1}{2}$

2.2 / Question 37

Given the equation

$$64x^2 + bx + 25 = 0$$

For which of the following values of b will the equation have more than one real solution?

A. -91

B. -80

C. 5

D. 40

Worked solution

2.2 / Q37

Step 1. Require a positive discriminant.

More than one real solution means two distinct roots.

$$\Delta = b^2 - 4(64)(25) = b^2 - 6400 > 0.$$

Step 2. Solve the condition.

$b^2 > 80^2$, equivalently $|b| > 80$.

Step 3. Check the choices.

Only $b = -91$ has absolute value greater than 80. At $b = -80$ the discriminant is zero, so it gives only one distinct root.

Final answer: A; -91

2.2 / Question 38

Given the system:

$$\begin{cases} y = -1.5 \\ y = x^2 + 8x + a \end{cases}$$

If the system has exactly one real solution, what is the value of a ?

Worked solution

2.2 / Q38

Step 1. Complete the square for the parabola.

$$y = x^2 + 8x + a = (x + 4)^2 + a - 16.$$

Step 2. Make the horizontal line tangent.

A horizontal line meets this upward-opening parabola once only at its minimum. Thus $a - 16 = -1.5$.

Step 3. Solve and check.

$a = 14.5 = 29/2$. The unique point is $(-4, -1.5)$.

Final answer: $\frac{29}{2} = 14.5$

2.2 / Question 39

In the xy -plane, the graph of $y = -x^2 + 9x - 100$ intersects the line $y = c$ at exactly one point. What is the value of c ?

A. $-\frac{481}{4}$

B. -100

C. $-\frac{319}{4}$

D. $-\frac{9}{2}$

Worked solution

2.2 / Q39

Step 1. Locate the vertex.

For $y = -x^2 + 9x - 100$,

$$x_v = -\frac{9}{2(-1)} = \frac{9}{2}.$$

Step 2. Compute the vertex height.

$$y_v = -\left(\frac{9}{2}\right)^2 + 9\left(\frac{9}{2}\right) - 100 = -\frac{319}{4}.$$

Step 3. Choose the tangent horizontal line.

Exactly one intersection occurs at the vertex, so $c = -319/4$.

Final answer: C; $-\frac{319}{4}$

2.2 / Question 40

Given the system:

$$\begin{cases} x^2 + y + 7 = 7 \\ 20x + 100 - y = 0 \end{cases}$$

The solution is (x, y) . What is the value of x ?

Worked solution

2.2 / Q40

Step 1. Solve both equations for y .

$$x^2 + y + 7 = 7 \Rightarrow y = -x^2, \quad 20x + 100 - y = 0 \Rightarrow y = 20x + 100.$$

Step 2. Equate and factor.

$$x^2 + 20x + 100 = (x + 10)^2 = 0.$$

Step 3. Find and verify the solution.

$x = -10$, and $y = -100$. Both original equations hold, so the requested value is -10 .

Final answer: -10

2.3 Nonlinear Functions

2.3 / Question 1

The quadratic function h is defined as $h(x) = 2(x - 4)^2 - 32$. In the xy -plane, the graph of $y = h(x)$ intersects the x -axis at the points $(0, 0)$ and $(t, 0)$, where t is a constant. What is the value of t ?

A. 1

B. 2

C. 4

D. 8

Worked solution

2.3 / Q1

Step 1. Set the height equal to zero.

At an x -intercept,

$$2(x - 4)^2 - 32 = 0 \quad \Rightarrow \quad (x - 4)^2 = 16.$$

Step 2. Solve both branches.

$x - 4 = \pm 4$, so $x = 0$ or $x = 8$.

Step 3. Identify the second intercept.

The first is $(0, 0)$, so the other is $(8, 0)$ and $t = 8$.

Final answer: D; 8

2.3 / Question 2

The function f is defined by $f(x) = (-8)(2)^x + 22$. What is the y -intercept of the graph of $y = f(x)$ in the xy -plane?

A. $(0, 14)$

B. $(0, 2)$

C. $(0, 22)$

D. $(0, -8)$

Worked solution

2.3 / Q2

Step 1. Use the definition of a y -intercept.

Set $x = 0$ in the exponential function $f(x) = -8 \cdot 2^x + 22$.

Step 2. Evaluate the zero power.

$$f(0) = -8(2^0) + 22 = -8 + 22 = 14.$$

Step 3. Write the coordinate pair.

The y -intercept is $(0, 14)$, not just the number 14.

Final answer: A; $(0, 14)$

2.3 / Question 3

The function f is defined by $f(x) = 9,000(0.66)^x$. The function models the number of advertisements a company sent to its clients each year, where x is the number of years since 1997, and $0 \leq x \leq 5$. If $y = f(x)$ is graphed in the xy -plane, which of the following is the best interpretation of the y -intercept?

- A. The minimum estimated number of advertisements the company sent to its clients during the 5 years was 1,708.
- B. The minimum estimated number of advertisements the company sent to its clients during the 5 years was 9,000.
- C. The estimated number of advertisements the company sent to its clients in 1997 was 1,708.
- D. The estimated number of advertisements the company sent to its clients in 1997 was 9,000.

Worked solution

2.3 / Q3

Step 1. Evaluate the initial output.

$$f(0) = 9000(0.66)^0 = 9000.$$

Step 2. Translate the input into a year.

The input counts years since 1997, so $x = 0$ means 1997.

Step 3. Interpret the intercept.

The model estimates that the company sent 9,000 advertisements in 1997. Because the model decreases, this is the initial value, not the minimum over the interval.

Final answer: D; 9,000 advertisements in 1997

2.3 / Question 4

The first term of a sequence is 9. Each term after the first is 4 times the preceding term. If w represents the n th term of the sequence, which equation gives w in terms of n ?

A. $w = 4(9^n)$

B. $w = 4(9^{n-1})$

C. $w = 9(4^n)$

D. $w = 9(4^{n-1})$

Worked solution

2.3 / Q4

Step 1. Identify the geometric-sequence rule.

The first term is 9 and the common ratio is 4.

Step 2. Count the multiplications.

To reach term n from term 1, multiply by 4 exactly $n - 1$ times.

$$w = 9 \cdot 4^{n-1}.$$

Step 3. Check the first two terms.

At $n = 1$, $w = 9$; at $n = 2$, $w = 36$, four times the first term.

Final answer: D; $w = 9 \cdot 4^{n-1}$

2.3 / Question 5

Function f is defined by $f(x) = -a^x + b$, where a and b are constants. In the xy -plane, the graph of $y = f(x) - 12$ has a y -intercept at $(0, -\frac{75}{7})$. The product of a and b is $\frac{320}{7}$. What is the value of a ?

Worked solution

2.3 / Q5

Step 1. Evaluate the shifted function at zero.

Read $f(x) = -a^x + b$ as an exponential function. Since $a^0 = 1$,

$$f(0) - 12 = -1 + b - 12 = b - 13.$$

Step 2. Use the given y -intercept.

$$b - 13 = -\frac{75}{7} \quad \Rightarrow \quad b = \frac{16}{7}.$$

Step 3. Use the product and solve for a .

$$ab = \frac{320}{7} \quad \Rightarrow \quad a = \frac{320/7}{16/7} = 20.$$

Final answer: 20

2.3 / Question 6

A machine launches a softball from ground level. The softball reaches a maximum height of 51.84 meters above the ground at 1.8 seconds and hits the ground at 3.6 seconds. Which equation represents the height above ground h , in meters, of the softball t seconds after it is launched?

A. $h = -t^2 + 3.6$

B. $h = -t^2 + 51.84$

C. $h = -16(t - 1.8)^2 - 3.6$

D. $h = -16(t - 1.8)^2 + 51.84$

Worked solution

2.3 / Q6

Step 1. Use vertex form.

The maximum occurs at $(1.8, 51.84)$, so

$$h = A(t - 1.8)^2 + 51.84, \quad A < 0.$$

Step 2. Use the launch height.

At $t = 0$, $h = 0$. Thus $0 = A(1.8)^2 + 51.84$, giving $A = -51.84/3.24 = -16$.

Step 3. Write and check the model.

$$h = -16(t - 1.8)^2 + 51.84.$$

At $t = 3.6$, this also gives $h = 0$, matching the landing time.

Final answer: D; $h = -16(t - 1.8)^2 + 51.84$

2.3 / Question 7

The function f is defined by $f(x) = a^x + b$, where a and b are constants. In the xy -plane, the graph of $y = f(x)$ has an x -intercept at $(2, 0)$ and a y -intercept at $(0, -323)$. What is the value of b ?

Worked solution

2.3 / Q7

Step 1. Evaluate the exponential at zero.

The function is $f(x) = a^x + b$, so

$$f(0) = 1 + b.$$

Step 2. Use the given intercept.

$$1 + b = -323 \quad \Rightarrow \quad b = -324.$$

Step 3. Check the other intercept.

$f(2) = 0$ requires $a^2 = 324$. The positive exponential base $a = 18$ works, confirming consistency.

Final answer: -324

2.3 / Question 8

The revenue $f(x)$, in dollars, that a company receives from sales of a product is given by the function $f(x) = -500x^2 + 25,000x$, where x is the unit price of the product in dollars. The graph of $y = f(x)$ intersects the x -axis at 0 and a . What does a represent?

- A. The revenue, in dollars, when the unit price of the product is \$0
- B. The unit price, in dollars, of the product that will result in maximum revenue
- C. The unit price, in dollars, of the product that will result in a revenue of \$0
- D. The maximum revenue, in dollars, that the company can make

Worked solution

2.3 / Q8

Step 1. Find the nonzero root.

$$f(x) = -500x^2 + 25,000x = -500x(x - 50).$$

Thus the nonzero x -intercept is $a = 50$.

Step 2. Interpret input and output.

The input x is the unit price; the output $f(x)$ is revenue. At an x -intercept, revenue is zero.

Step 3. State the meaning.

a represents the unit price that produces \$0 revenue. It does not represent the price maximizing revenue, which lies midway between the roots.

Final answer: C; the unit price producing zero revenue

2.3 / Question 9

Growth of a Culture of Bacteria

Day	Number of bacteria per milliliter at end of day
1	2.5×10^5
2	5.0×10^5
3	1.0×10^6

A culture of bacteria is growing at an exponential rate, as shown in the table above. At this rate, on which day would the number of bacteria per milliliter reach 5.12×10^8 ?

A. Day 5

B. Day 9

C. Day 11

D. Day 12

Worked solution

2.3 / Q9

Step 1. Identify the daily growth factor.

The table doubles each day: $5.0 \times 10^5 / (2.5 \times 10^5) = 2$.

Step 2. Write the model from day 1.

$$N(d) = 2.5 \times 10^5 \cdot 2^{d-1}.$$

Step 3. Solve for the day.

$$2^{d-1} = \frac{5.12 \times 10^8}{2.5 \times 10^5} = 2048 = 2^{11}.$$

Therefore $d - 1 = 11$ and $d = 12$.

Final answer: D; Day 12

2.3 / Question 10

The function $f(x) = ax^2 + 4x + c$ is a quadratic function, where a and c are constants. The graph of $y = f(x)$ in the xy -plane is a parabola that opens upward and has a vertex at point (h, k) . If $k < 0$ and $f(-9) = f(3)$, which of the following must be true?

I. $c < 0$

II. $a \geq 1$

A. I only

B. II only

C. I and II

D. Neither I nor II

Worked solution

2.3 / Q10

Step 1. Find the axis of symmetry.

Since $f(-9) = f(3)$, the axis is halfway between these inputs:

$$h = \frac{-9 + 3}{2} = -3.$$

Step 2. Determine a .

For $f(x) = ax^2 + 4x + c$, $h = -4/(2a) = -2/a$. Thus $a = 2/3$, so statement II ($a \geq 1$) is false.

Step 3. Test whether statement I is necessary.

At the vertex, $k = c - 6 < 0$, giving only $c < 6$. For example, $c = 1$ gives $k = -5 < 0$ while $c > 0$. Thus I need not be true either.

Final answer: D; neither I nor II

2.3 / Question 11

The function $f(x) = 4x^2 + 64x + 262$ is defined, and $g(x) = f(x + 5)$. For what value of x does $g(x)$ reach its minimum?

A. -13

B. -8

C. -5

D. -3

Worked solution

2.3 / Q11

Step 1. Find the minimum input for f .

$$x_v = -\frac{64}{2(4)} = -8.$$

The positive leading coefficient guarantees a minimum.

Step 2. Match the input of g to that value.

Since $g(x) = f(x + 5)$, its minimum occurs when $x + 5 = -8$.

Step 3. Solve.

$x = -13$. Equivalently, $f(x) = 4(x + 8)^2 + 6$, so $g(x) = 4(x + 13)^2 + 6$.

Final answer: A; -13

2.3 / Question 12

The function $f(x) = (x - 6)(x - 2)(x + 6)$ is defined. In the xy -plane, the graph of $y = g(x)$ is the result of translating the graph of $y = f(x)$ up 4 units. What is the value of $g(0)$?

Worked solution

2.3 / Q12

Step 1. Express the upward shift.

Translating up 4 units gives $g(x) = f(x) + 4$.

Step 2. Evaluate the original function at zero.

$$f(0) = (-6)(-2)(6) = 72.$$

Step 3. Add the vertical shift.

$$g(0) = 72 + 4 = 76.$$

Final answer: 76

2.3 / Question 13

A quadratic function models a projectile's height, in meters, above the ground in terms of the time, in seconds, after it was launched. The model estimates that the projectile was launched from an initial height of 7 meters above the ground and reached a maximum height of 51.1 meters above the ground 3 seconds after the launch. How many seconds after the launch does the model estimate that the projectile will return to a height of 7 meters?

A. 3

B. 6

C. 7

D. 9

Worked solution

2.3 / Q13

Step 1. Use the parabola's symmetry.

The maximum occurs at $t = 3$, so the axis of symmetry is $t = 3$.

Step 2. Reflect the initial time across that axis.

The initial height of 7 meters occurs at $t = 0$, which is 3 seconds before the vertex. The same height occurs 3 seconds after it.

Step 3. Calculate and check the later time.

The time is $t = 6$. Indeed, $h(t) = -4.9(t - 3)^2 + 51.1$ gives $h(0) = h(6) = 7$.

Final answer: B; 6 seconds

2.3 / Question 14

Given $y = x^2 - 14x + 22$, for what value of x does the value of y reach its minimum?

Worked solution

2.3 / Q14

Step 1. Identify an upward-opening quadratic.

The coefficient of x^2 is positive, so the vertex gives the minimum.

Step 2. Find its x -coordinate.

$$x_v = -\frac{-14}{2(1)} = 7.$$

Step 3. Confirm by completing the square.

$y = (x - 7)^2 - 27$, minimized when $x = 7$. The question asks for the input, not the minimum output -27 .

Final answer: 7

2.3 / Question 15

Function f is defined by $f(x) = -a^x + b$, where a and b are constants. In the xy -plane, the graph of $y = f(x) - 15$ has a y -intercept at $\left(0, -\frac{99}{7}\right)$. The product of a and b is $\frac{65}{7}$. What is the value of a ?

Worked solution

2.3 / Q15

Step 1. Evaluate the shifted function at zero.Because $f(x) = -a^x + b$,

$$f(0) - 15 = -1 + b - 15 = b - 16.$$

Step 2. Use the y -intercept to find b .

$$b - 16 = -\frac{99}{7} \quad \Rightarrow \quad b = \frac{13}{7}.$$

Step 3. Solve using the product.

$$ab = \frac{65}{7} \quad \Rightarrow \quad a = \frac{65/7}{13/7} = 5.$$

Final answer: 5

2.3 / Question 16

The function $f(x) = (x - 14)(x + 19)$ is defined. For what value of x does $f(x)$ reach its minimum?

A. -266

B. -19

C. $-\frac{33}{2}$

D. $-\frac{5}{2}$

Worked solution

2.3 / Q16

Step 1. Read the roots.

$f(x) = (x - 14)(x + 19)$ has roots 14 and -19 .

Step 2. Find their midpoint.

The vertex lies halfway between the roots:

$$x_v = \frac{14 + (-19)}{2} = -\frac{5}{2}.$$

Step 3. Confirm it is a minimum.

The leading coefficient is 1, so the parabola opens upward. Its minimum occurs at $x = -5/2$.

Final answer: D; $-\frac{5}{2}$

2.3 / Question 17

A model estimates that from 2015 to 2020, the number of squirrels in a population increased by 150% of the previous year's value. If at the end of 2016, there were 180 squirrels, which equation represents this model, where n is the estimated number of squirrels t years after 2015 and $t \leq 5$?

A. $n = 72(1.5)^t$

B. $n = 72(2.5)^t$

C. $n = 180(1.5)^t$

D. $n = 180(2.5)^t$

Worked solution

2.3 / Q17

Step 1. Convert the percentage increase to a multiplier.

An increase of 150% means the new value is $100\% + 150\% = 250\%$ of the old value, so the growth factor is 2.5.

Step 2. Find the initial population.

The end of 2016 corresponds to $t = 1$. If $n = A(2.5)^t$, then $180 = 2.5A$, so $A = 72$.

Step 3. Write and check the model.

$$n = 72(2.5)^t.$$

At $t = 1$, this gives 180, as required.

Final answer: B; $n = 72(2.5)^t$

2.3 / Question 18

The function $f(x) = |59 - 2x|$ is defined. For which of the following values of k does $f(k) = 3k$?

A. $\frac{59}{5}$

B. $\frac{59}{2}$

C. $\frac{177}{5}$

D. 59

Worked solution

2.3 / Q18

Step 1. Use the sign restriction.

$|59 - 2k| = 3k$ requires $k \geq 0$ because an absolute value is nonnegative.

Step 2. Solve the two branches.

$59 - 2k = 3k$ gives $k = 59/5$. The other branch, $59 - 2k = -3k$, gives $k = -59$, which violates $k \geq 0$.

Step 3. Check the valid result.

At $k = 59/5$, $|59 - 118/5| = 177/5 = 3k$.

Final answer: A; $\frac{59}{5}$

2.3 / Question 19

The population P of a certain city y years after the last census is modeled by the equation $P = P_0(1+r)^y$, where r is a constant and P_0 is the population when $y = 0$. If the population decreases each year by a fixed percent, which of the following must be true?

A. $r < -1$

B. $-1 < r < 0$

C. $0 < r < 1$

D. $r > 1$

Worked solution

2.3 / Q19

Step 1. Interpret the annual factor.

The population is multiplied by $1 + r$ each year.

Step 2. Require positive decay.

For a positive population to decrease by a fixed percentage each year, its multiplier must satisfy

$$0 < 1 + r < 1.$$

Step 3. Solve for r .

Subtract 1 throughout to obtain $-1 < r < 0$.

Final answer: B; $-1 < r < 0$

2.3 / Question 20

The function $f(x) = 5,470(0.64)^{\frac{x}{12}}$ gives the value, in dollars, of a certain piece of equipment after x months of use. If the value decreases each year by $p\%$, what is the value of p ?

A. 4

B. 5

C. 36

D. 64

Worked solution

2.3 / Q20

Step 1. Convert a year into the model's time unit.

A year is 12 months, so increasing x by 12 increases the exponent $x/12$ by 1.

Step 2. Find the yearly multiplier.

The equipment retains $0.64 = 64\%$ of its value each year.

Step 3. Find the percentage decrease.

$$p = 100 - 64 = 36.$$

Final answer: C; 36

2.3 / Question 21

The function $f(x) = \frac{1}{9}(x - 7)^2 + 3$ gives a metal ball's height above the ground $f(x)$, in inches, x seconds after it started moving, where $0 \leq x \leq 10$. What is the best interpretation of the vertex of the graph $y = f(x)$?

- A. The metal ball's minimum height was 3 inches above the ground.
- B. The metal ball's minimum height was 7 inches above the ground.
- C. The metal ball's height was 3 inches above the ground when it started moving.
- D. The metal ball's height was 7 inches above the ground when it started moving.

Worked solution

2.3 / Q21

Step 1. Read the vertex from vertex form.

$f(x) = \frac{1}{9}(x - 7)^2 + 3$ has vertex $(7, 3)$.

Step 2. Check the direction and domain.

The coefficient $1/9$ is positive, and $x = 7$ lies within $0 \leq x \leq 10$. Thus the vertex is the minimum on the stated interval.

Step 3. Interpret the output.

The ball's minimum height is 3 inches, reached after 7 seconds.

Final answer: A; minimum height of 3 inches

2.3 / Question 22

The function $f(t) = 8,000(0.65)^t$ models the number of coupons a company sent to customers at the end of each year, where t is the number of years since the end of 1998 and $0 \leq t \leq 5$. What is the best interpretation of the y -intercept of the graph of $y = f(t)$?

- A. The minimum estimated number of coupons the company sent to their customers during the 5 years was 1,428.
- B. The minimum estimated number of coupons the company sent to their customers during the 5 years was 8,000.
- C. The estimated number of coupons the company sent to their customers at the end of 1998 was 1,428.
- D. The estimated number of coupons the company sent to their customers at the end of 1998 was 8,000.

Worked solution

2.3 / Q22

Step 1. Evaluate the y -intercept.

$$f(0) = 8000(0.65)^0 = 8000.$$

Step 2. Interpret the time origin.

$t = 0$ corresponds to the end of 1998.

Step 3. State the contextual meaning.

The model estimates 8,000 coupons sent at the end of 1998. This is the initial value, rather than the minimum over the five-year interval.

Final answer: D; 8,000 coupons at the end of 1998

2.3 / Question 23

In the xy -plane, a parabola has vertex $(9, -14)$ and intersects the x -axis at two points. If the equation of the parabola is written as $y = ax^2 + bx + c$, what could be the value of $a + b + c$?

A. -23

B. -19

C. -14

D. -12

Worked solution

2.3 / Q23

Step 1. Use the vertex and intercept information.

Write $y = a(x - 9)^2 - 14$. With a vertex below the x -axis and two x -intercepts, the parabola must open upward, so $a > 0$.

Step 2. Relate the requested sum to an output.

In standard form, $a + b + c = f(1)$. Therefore

$$f(1) = a(1 - 9)^2 - 14 = 64a - 14 > -14.$$

Step 3. Select a possible value.

Only -12 is greater than -14 . It is attainable with $a = 1/32 > 0$, which also gives two x -intercepts.

Final answer: D; -12

2.3 / Question 24

The table below gives selected values of a polynomial function p :

x	$p(x)$
-2	5
-1	0
0	-3
1	-1
2	0

Based on the values in the table, which of the following must be a factor of p ?

A. $(x - 3)$

B. $(x + 3)$

C. $(x - 1)(x + 2)$

D. $(x + 1)(x - 2)$

Worked solution

2.3 / Q24

Step 1. Identify the zeros in the table.

The table gives $p(-1) = 0$ and $p(2) = 0$.

Step 2. Apply the factor theorem.

Thus $x + 1$ and $x - 2$ are factors of the polynomial p .

Step 3. Combine the distinct factors.

Because the roots are distinct, $(x + 1)(x - 2)$ must divide $p(x)$. This does not require p to be quadratic.

Final answer: D; $(x + 1)(x - 2)$

2.3 / Question 25

When the quadratic function f is graphed in the xy -plane, where $y = f(x)$, its vertex is $(-3, 6)$. One of the x -intercepts of this graph is $\left(-\frac{17}{4}, 0\right)$. What is the other x -intercept?

A. $\left(-\frac{29}{4}, 0\right)$

B. $\left(-\frac{7}{4}, 0\right)$

C. $\left(\frac{5}{4}, 0\right)$

D. $\left(\frac{17}{4}, 0\right)$

Worked solution

2.3 / Q25

Step 1. Use symmetry about the vertex.

The axis is $x = -3$, so the two roots have average -3 .

Step 2. Solve for the other root.

Let it be r . Then

$$\frac{-17/4 + r}{2} = -3 \quad \Rightarrow \quad r = -6 + \frac{17}{4} = -\frac{7}{4}.$$

Step 3. Write the intercept.

The other x -intercept is $(-7/4, 0)$. Both roots are $5/4$ unit from the axis.

Final answer: B; $\left(-\frac{7}{4}, 0\right)$

2.3 / Question 26

The function $f(x) = (x + 3)(x + 1)$ is graphed in the xy -plane. Which of the following intervals contains the x -coordinate of the vertex of the graph of f ?

A. $-4 < x < -3$

B. $-3 < x < 1$

C. $1 < x < 3$

D. $3 < x < 4$

Worked solution

2.3 / Q26

Step 1. Read the zeros.

The roots of $(x + 3)(x + 1)$ are -3 and -1 .

Step 2. Find the vertex's x -coordinate.

$$x_v = \frac{-3 + (-1)}{2} = -2.$$

Step 3. Locate that value in the choices.

$-3 < -2 < 1$, so the interval in B contains the vertex's input.

Final answer: B; $-3 < x < 1$

2.3 / Question 27

A landscaper is designing a rectangular garden. The length is 5 feet longer than the width. If the area is 104 square feet, what will be the length in feet?

Worked solution

2.3 / Q27

Step 1. Define the dimensions.

Let the width be $w > 0$ feet. The length is $w + 5$ feet, and

$$w(w + 5) = 104.$$

Step 2. Factor the area equation.

$$w^2 + 5w - 104 = (w + 13)(w - 8) = 0.$$

The positive width is $w = 8$.

Step 3. Find and verify the length.

The length is $8 + 5 = 13$ feet, and $8(13) = 104$ square feet.

Final answer: 13 feet

2.3 / Question 28

$P(t) = 290(1.04)^{\frac{4}{6}t}$ models the population, in thousands, of a certain city t years after 2005. The population increases by $n\%$ every 18 months. What is the value of n ?

A. 0.38

B. 1.04

C. 4

D. 6

Worked solution

2.3 / Q28

Step 1. Convert the growth interval to years.

Eighteen months is $18/12 = 3/2$ years.

Step 2. Find the change in exponent.

In $P(t) = 290(1.04)^{(4/6)t}$, a $3/2$ -year increase changes the exponent by

$$\frac{4}{6} \cdot \frac{3}{2} = 1.$$

Step 3. Interpret the factor.

Over 18 months, the population is multiplied by 1.04, an increase of 4%. Thus $n = 4$.

Final answer: C; 4

2.3 / Question 29

For the function f , $f(0) = 86$, and for each increase in x by 1, $f(x)$ decreases by 80%. What is the value of $f(2)$?

Worked solution

2.3 / Q29

Step 1. Find the retained proportion.

A decrease of 80% leaves $1 - 0.80 = 0.20$ of the previous value per unit increase in x .

Step 2. Apply the factor twice.

$$f(2) = 86(0.20)^2.$$

Step 3. Calculate.

$86(0.04) = 3.44 = 86/25$. Check: $f(1) = 17.2$ and $0.2(17.2) = 3.44$.

Final answer: $\frac{86}{25} = 3.44$

2.3 / Question 30

$M = 1,800(1.02)^t$ models gym members, M , of a gym t years after opening. Which equation models the number of members q quarter-years after opening?

A. $M = 1,800(1.02)^{q/4}$

B. $M = 1,800(1.02)^{4q}$

C. $M = 1,800(1.005)^{4q}$

D. $M = 1,800(1.082)^q$

Worked solution

2.3 / Q30

Step 1. Relate quarter-years to years.

There are four quarter-years per year, so $t = q/4$.

Step 2. Substitute into the original model.

$$M = 1800(1.02)^{q/4}.$$

Step 3. Check a full year.

At $q = 4$, the model gives $1800(1.02)$, exactly one year's growth.

Final answer: A; $M = 1800(1.02)^{q/4}$

2.3 / Question 31

The functions f and g are defined by the given equation, where $x \geq 0$. Which function shows a maximum value as a constant or coefficient for $x \geq 0$?

I $f(x) = 33(0.4)^{x+3}$

II $g(x) = 33(0.16)(0.4)^{x-2}$.

A. I only

B. II only

C. I and II

D. Neither I nor II

Worked solution

2.3 / Q31

Step 1. Locate the maxima on the stated domain.

Since $0 < 0.4 < 1$, both positive exponential functions decrease as x increases. On $x \geq 0$, each maximum occurs at $x = 0$.

Step 2. Evaluate function I.

$$f(0) = 33(0.4)^3 = 2.112.$$

This maximum is not a displayed constant or coefficient in form I.

Step 3. Evaluate function II.

$$g(0) = 33(0.16)(0.4)^{-2} = 33 \frac{0.16}{0.16} = 33.$$

The number 33 appears explicitly as a coefficient in II. Thus II only shows its maximum in the requested way.

Final answer: B; II only

2.3 / Question 32

Square P has a side length of x inches. Square Q has a perimeter that is 176 inches greater than the perimeter of square P . The function f gives the area of square Q , in inches. Which of the following defines f ?

A. $f(x) = (x + 44)^2$

B. $f(x) = (x + 176)^2$

C. $f(x) = (176x + 44)^2$

D. $f(x) = (176x + 176)^2$

Worked solution

2.3 / Q32

Step 1. Find the perimeter of square Q.

Square P has perimeter $4x$, so Q has perimeter $4x + 176$ inches.

Step 2. Find Q's side length.

$$s_Q = \frac{4x + 176}{4} = x + 44.$$

Step 3. Square the side length.

$$f(x) = (x + 44)^2.$$

Area is measured in *square inches*; the source's wording "in inches" is a unit typo.

Final answer: A; $f(x) = (x + 44)^2$

2.3 / Question 33

The surface area of a cube is $6\left(\frac{a}{4}\right)^2$, where a is a positive constant. Which of the following gives the perimeter of one face of the cube?

A. $\frac{a}{4}$

B. a

C. $4a$

D. $6a$

Worked solution

2.3 / Q33

Step 1. Use the cube surface-area formula.

For side length $s > 0$, the surface area is $6s^2$. Thus

$$6s^2 = 6\left(\frac{a}{4}\right)^2.$$

Step 2. Find the side length.

Since $a > 0$, $s = a/4$.

Step 3. Calculate one face's perimeter.

Each face is a square, so its perimeter is $4s = 4(a/4) = a$.

Final answer: B; a

2.3 / Question 34

The function $h(x) = a^x + b$, where a and b are positive constants, passes through $(0, 10)$ and $(-2, \frac{325}{36})$. What is the value of ab ?

A. $\frac{1}{4}$

B. $\frac{1}{2}$

C. 54

D. 60

Worked solution

2.3 / Q34

Step 1. Use the point with input zero.

The function is exponential: $h(x) = a^x + b$. Thus $h(0) = 1 + b = 10$, giving $b = 9$.

Step 2. Use the second point.

$$a^{-2} + 9 = \frac{325}{36} \Rightarrow a^{-2} = \frac{1}{36}.$$

Hence $a^2 = 36$, and positivity gives $a = 6$.

Step 3. Calculate the product.

$$ab = 6(9) = 54.$$

Final answer: C; 54

2.3 / Question 35

Table:

x	y
21	-8
23	8
25	-8

The table shows three values of x and their corresponding values of y . Given $y = f(x) + 4$, and f is quadratic function, what is the y -coordinate of the y -intercept of $y = f(x)$?

Worked solution

2.3 / Q35

Step 1. Recover the quadratic values of f .

The table gives $y = f(x) + 4$, so subtract 4: $f(21) = -12$, $f(23) = 4$, and $f(25) = -12$. Equal values at 21 and 25 place the axis at $x = 23$.

Step 2. Write and determine the vertex form.

$$f(x) = A(x - 23)^2 + 4.$$

Using $f(21) = -12$ gives $4A + 4 = -12$, so $A = -4$.

Step 3. Evaluate the y -intercept.

$$f(0) = -4(23^2) + 4 = -2116 + 4 = -2112.$$

Final answer: -2112

2.3 / Question 36

A quadratic function models the height, in feet, of an object above the ground in terms of the time, in seconds, after the object is launched off an elevated surface. The model indicates the object has an initial height of 10 feet above the ground and reaches its maximum height of 1,034 feet above the ground 8 seconds after being launched. Based on the model, what is the height, in feet, of the object above the ground 10 seconds after being launched?

A. 234

B. 778

C. 970

D. 1,014

Worked solution

2.3 / Q36

Step 1. Write vertex form.

The maximum is 1,034 feet at $t = 8$, so

$$h(t) = A(t - 8)^2 + 1034.$$

Step 2. Use the initial height.

$h(0) = 10$ gives $64A + 1034 = 10$, so $A = -16$.

Step 3. Evaluate at 10 seconds.

$$h(10) = -16(10 - 8)^2 + 1034 = -64 + 1034 = 970.$$

Final answer: C; 970 feet

2.3 / Question 37

Function f is defined by $f(x) = (x + 6)(x + 5)(x + 1)$. Function g is defined by $g(x) = f(x - 1)$. The graph of $y = g(x)$ in the xy -plane has x -intercepts at $(a, 0)$, $(b, 0)$, and $(c, 0)$, where a , b , and c are distinct constants. What is the value of $a + b + c$?

A. -15

B. -9

C. 11

D. 15

Worked solution

2.3 / Q37

Step 1. Substitute $x - 1$ into every factor.

$$g(x) = f(x - 1) = (x + 5)(x + 4)x.$$

Step 2. Find the zeros.

The three distinct roots are -5 , -4 , and 0 .

Step 3. Add the intercept inputs.

$$a + b + c = -5 - 4 + 0 = -9.$$

Final answer: B; -9

2.3 / Question 38

The total distance d , in meters, traveled by an object moving in a straight line can be modeled by a quadratic function that is defined in terms of t , where t is the time in seconds. At a time of 10.0 seconds, the total distance traveled by the object is 50.0 meters, and at a time of 20.0 seconds, the total distance traveled by the object is 200.0 meters. If the object was at a distance of 0 meters when $t = 0$, then what is the total distance traveled, in meters, by the object after 30.0 seconds?

Worked solution

2.3 / Q38

Step 1. Set up the quadratic model.

Let $d(t) = At^2 + Bt + C$. Since $d(0) = 0$, $C = 0$. The other data give

$$100A + 10B = 50, \quad 400A + 20B = 200.$$

Step 2. Solve for the coefficients.

Subtract twice the first equation from the second: $200A = 100$, so $A = 1/2$. Then $50 + 10B = 50$ gives $B = 0$.

Step 3. Evaluate the requested time.

$$d(30) = \frac{1}{2}(30)^2 = 450 \text{ meters.}$$

Final answer: 450 meters

2.3 / Question 39

$f(x) = 4x^2 - 50x + 126$ The given equation defines the function f . For what value of x does $f(x)$ reach its minimum?

Worked solution

2.3 / Q39

Step 1. Identify the coefficients.

For $f(x) = 4x^2 - 50x + 126$, the leading coefficient is positive, so the vertex is a minimum.

Step 2. Find the vertex input.

$$x_v = -\frac{-50}{2(4)} = \frac{50}{8} = \frac{25}{4}.$$

Step 3. Express the value.

The minimum occurs at $x = 25/4 = 6.25$. The question asks for the input, not the minimum output.

Final answer: $\frac{25}{4} = 6.25$

2.3 / Question 40

$$h(x) = x^3 + ax^2 + bx + c$$

The function h is defined above, where a , b , and c are integer constants. If the zeros of the function are -5 , 6 , and 7 , what is the value of c ?

Worked solution

2.3 / Q40

Step 1. Write the monic polynomial from its zeros.

The coefficient of x^3 is 1, so

$$h(x) = (x + 5)(x - 6)(x - 7).$$

Step 2. Evaluate the constant term.

The constant is $h(0)$:

$$c = (5)(-6)(-7).$$

Step 3. Calculate and check the sign.

$c = 210$. Two negative factors make the product positive.

Final answer: 210

2.3 / Question 41

$$f(x) = 9(4)^x$$

The function f is defined by the given equation. If $g(x) = f(x+2)$, which of the following equations defines the function g ?

A. $g(x) = 18(4)^x$

B. $g(x) = 144(4)^x$

C. $g(x) = 18(8)^x$

D. $g(x) = 81(16)^x$

Worked solution

2.3 / Q41

Step 1. Substitute the shifted input.

$$g(x) = f(x+2) = 9 \cdot 4^{x+2}.$$

Step 2. Separate the powers.

$$9 \cdot 4^{x+2} = 9 \cdot 4^x \cdot 4^2.$$

Step 3. Simplify the coefficient.

$9(16) = 144$, so $g(x) = 144 \cdot 4^x$. Check: $g(0) = f(2) = 144$.

Final answer: B; $g(x) = 144 \cdot 4^x$

2.3 / Question 42

$$f(x) = (x + 7)^2 + 4$$

The function f is defined by the given equation. For what value of x does $f(x)$ reach its minimum?

Worked solution

2.3 / Q42

Step 1. Recognize the nonnegative term.

In $f(x) = (x + 7)^2 + 4$, the square is always at least zero.

Step 2. Make that term as small as possible.

The square is zero when $x + 7 = 0$.

Step 3. State the minimizing input.

$x = -7$, where $f(-7) = 4$. The requested answer is the input -7 .

Final answer: -7

2.3 / Question 43

Kao measured the temperature of a cup of hot chocolate placed in a room with a constant temperature of 70 degrees Fahrenheit ($^{\circ}\text{F}$). The temperature of the hot chocolate was 185°F at 6:00 p.m. when it started cooling. The temperature of the hot chocolate was 156°F at 6:05 p.m. and 135°F at 6:10 p.m. The hot chocolate's temperature continued to decrease. Of the following functions, which best models the temperature $T(m)$, in degrees Fahrenheit, of Kao's hot chocolate m minutes after it started cooling?

A. $T(m) = 185(1.25)^m$

B. $T(m) = 185(0.85)^m$

C. $T(m) = (185 - 70)(0.75)^{\frac{m}{5}}$

D. $T(m) = 70 + 115(0.75)^{\frac{m}{5}}$

Worked solution

2.3 / Q43

Step 1. Model the excess temperature above room temperature.

At the start, the excess is $185 - 70 = 115$ degrees Fahrenheit. The model must start at 185 and approach 70 as time increases.

Step 2. Estimate the five-minute factor.

After 5 minutes the excess is 86; after 10 minutes it is 65. The ratios $86/115 \approx 0.748$ and $65/86 \approx 0.756$ are both close to 0.75.

Step 3. Write and check the model.

$$T(m) = 70 + 115(0.75)^{m/5}.$$

It predicts $T(0) = 185$, $T(5) = 156.25$, and $T(10) = 134.6875$, close to the measurements.

Final answer: D; $T(m) = 70 + 115(0.75)^{m/5}$

2.3 / Question 44

$$p(t) = 90,000(1.06)^t$$

The given function p models the population of Lowell t years after a census. Which of the following functions best models the population of Lowell m months after the census?

A. $r(m) = \frac{90,000}{12}(1.06)^m$

B. $r(m) = 90,000 \left(\frac{1.06}{12} \right)^m$

C. $r(m) = 90,000 \left(\frac{1.06}{12} \right)^{\frac{m}{12}}$

D. $r(m) = 90,000(1.06)^{\frac{m}{12}}$

Worked solution

2.3 / Q44

Step 1. Convert months to years.

At m months after the census, the elapsed time in years is $t = m/12$.

Step 2. Substitute into the given function.

$$r(m) = 90,000(1.06)^{m/12}.$$

Step 3. Check the starting and yearly values.

$r(0) = 90,000$ and $r(12) = 90,000(1.06)$, matching the original model.

Final answer: D; $r(m) = 90,000(1.06)^{m/12}$

2.3 / Question 45

The population of a town is currently 50,000, and the population is estimated to increase each year by 3% from the previous year. Which of the following equations can be used to estimate the number of years t it will take for the population of the town to reach 60,000?

A. $50,000 = 60,000(0.03)^t$

B. $50,000 = 60,000(3)^t$

C. $60,000 = 50,000(0.03)^t$

D. $60,000 = 50,000(1.03)^t$

Worked solution

2.3 / Q45

Step 1. Find the annual growth factor.

A 3% annual increase corresponds to multiplication by $1 + 0.03 = 1.03$.

Step 2. Write the population after t years.

$$P(t) = 50,000(1.03)^t.$$

Step 3. Set the model equal to the target.

$$60,000 = 50,000(1.03)^t.$$

The question asks for an equation, so solving for t is unnecessary.

Final answer: D; $60,000 = 50,000(1.03)^t$

2.3 / Question 46

For the exponential function f , the value of $f(1)$ is k , where k is a constant. Which of the following equivalent forms of the function f shows the value of k as the coefficient or the base?

A. $f(x) = 50(2)^{x+1}$

B. $f(x) = 80(2)^x$

C. $f(x) = 128(2)^{x-1}$

D. $f(x) = 205(2)^{x-2}$

Worked solution

2.3 / Q46

Step 1. Look for an exponent that becomes zero at $x = 1$.

For a form $A \cdot 2^{x-1}$, substituting $x = 1$ gives $A \cdot 2^0 = A$, displaying $f(1)$ as the coefficient.

Step 2. Evaluate choice C.

$$f(1) = 128 \cdot 2^{1-1} = 128.$$

Thus C has the requested structural feature.

Step 3. Check the source wording.

At $x = 1$, A, B, C, and D give 200, 160, 128, and 102.5, respectively. They are *not equivalent functions* as printed. C is the intended answer because only C directly displays its value at 1 as a coefficient; the source's "equivalent forms" claim is incorrect.

Final answer: C (intended answer; source wording inconsistent)

2.3 / Question 47

$$h(x) = -16x^2 + 100x + 10$$

The quadratic function above models the height above the ground h , in feet, of a projectile x seconds after it had been launched vertically. If $y = h(x)$ is graphed in the xy -plane, which of the following represents the real-life meaning of the positive x -intercept of the graph?

- A. The initial height of the projectile
- B. The maximum height of the projectile
- C. The time at which the projectile reaches its maximum height
- D. The time at which the projectile hits the ground

Worked solution

2.3 / Q47

Step 1. Interpret an x -intercept.

At an x -intercept, the output is zero, so $h(x) = 0$ means the projectile is at ground level.

Step 2. Interpret the positive input.

Here x measures seconds after launch. The positive root represents a time after the projectile was launched.

Step 3. State the physical event.

The positive x -intercept gives the time at which the projectile hits the ground. The initial height is instead $h(0) = 10$ feet.

Final answer: D; the time when the projectile hits the ground

2.3 / Question 48

x	$f(x)$
1	a
2	a^5
3	a^9

For the exponential function f , the table above shows several values of x and their corresponding values of $f(x)$, where a is a constant greater than 1. If k is a constant and $f(k) = a^{29}$, what is the value of k ?

Worked solution

2.3 / Q48

Step 1. Find the common ratio.

Successive outputs are multiplied by a^4 :

$$\frac{a^5}{a} = a^4, \quad \frac{a^9}{a^5} = a^4.$$

Step 2. Write the general function.

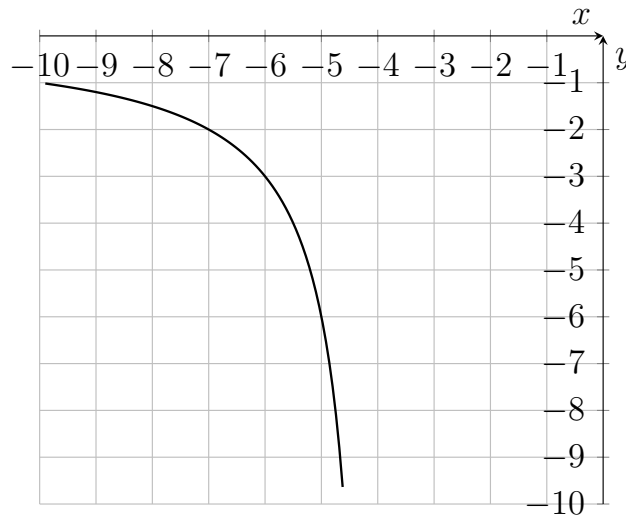
Starting from $f(1) = a$,

$$f(x) = a(a^4)^{x-1} = a^{4x-3}.$$

Step 3. Match the exponents.

Since $a > 1$, $a^{4k-3} = a^{29}$ implies $4k - 3 = 29$, so $k = 8$.

Final answer: 8

2.3 / Question 49

The rational function f is defined by an equation in the form $f(x) = \frac{a}{x+b}$, where a and b are constants. The partial graph of $y = f(x)$ is shown. If $g(x) = f(x+4)$, which equation could define function g ?

A. $g(x) = \frac{6}{x}$

B. $g(x) = \frac{6}{x+4}$

C. $g(x) = \frac{6}{x+8}$

D. $g(x) = \frac{6(x-4)}{x+4}$

Worked solution

2.3 / Q49

Step 1. Read points consistent with the graph.

The curve passes through approximately $(-10, -1)$ and $(-6, -3)$. For $f(x) = a/(x+b)$, these give $a = 10 - b$ and $a = 18 - 3b$.

Step 2. Find a rational function that fits.

Equating gives $b = 4$ and $a = 6$, so $f(x) = 6/(x + 4)$. This also predicts $f(-5) = -6$, consistent with the graph.

Step 3. Apply the horizontal input shift.

$$g(x) = f(x + 4) = \frac{6}{(x + 4) + 4} = \frac{6}{x + 8}, \quad x \neq -8.$$

The graph shifts left by 4 units.

Final answer: C; $g(x) = \frac{6}{x+8}$

2.3 / Question 50

$$f(x) = (x - 10)(x + 13)$$

The function f is defined by the given equation. For what value of x does $f(x)$ reach its minimum?

A. -130

B. -13

C. $-\frac{23}{2}$

D. $-\frac{3}{2}$

Worked solution

2.3 / Q50

Step 1. Read the roots.

For $f(x) = (x - 10)(x + 13)$, the roots are 10 and -13 .

Step 2. Find the axis of symmetry.

$$x_v = \frac{10 + (-13)}{2} = -\frac{3}{2}.$$

Step 3. Confirm the minimum.

The positive leading coefficient makes the vertex a minimum, so the minimizing input is $-3/2$.

Final answer: D; $-\frac{3}{2}$

2.3 / Question 51

For the function q , the value of $q(x)$ decreases by 45% for every increase in the value of x by 1. If $q(0) = 14$, which equation defines q ?

A. $q(x) = 0.55(14)^x$

B. $q(x) = 1.45(14)^x$

C. $q(x) = 14(0.55)^x$

D. $q(x) = 14(1.45)^x$

Worked solution

2.3 / Q51

Step 1. Convert the decrease to a multiplier.

A 45% decrease retains $100\% - 45\% = 55\%$, so the per-unit factor is 0.55.

Step 2. Use the initial value.

$q(0) = 14$ gives the exponential model

$$q(x) = 14(0.55)^x.$$

Step 3. Check both defining conditions.

$q(0) = 14$ and $q(x+1)/q(x) = 0.55$, so each step is a 45% decrease.

Final answer: C; $q(x) = 14(0.55)^x$

2.3 / Question 52

Two variables, x and y , are related such that for each increase of 1 in the value of x , the value of y increases by a factor of 4. When $x = 0$, $y = 200$. Which equation represents this relationship?

A. $y = 4(x)^{200}$

B. $y = 4(200)^x$

C. $y = 200(4)^x$

D. $y = 200(4x)$

Worked solution

2.3 / Q52

Step 1. Identify the exponential multiplier.

Each unit increase in x multiplies y by 4, so write $y = A4^x$.

Step 2. Use the initial condition.

At $x = 0$, $200 = A4^0 = A$.

Step 3. State and check the rule.

$$y = 200 \cdot 4^x.$$

Increasing x by 1 multiplies the output by 4, as required.

Final answer: C; $y = 200 \cdot 4^x$

2.3 / Question 53

What is the minimum value of the function f defined by $f(x) = (x - 2)^2 - 4$?

A. -4

B. -2

C. 2

D. 4

Worked solution

2.3 / Q53

Step 1. Use the nonnegativity of a square.

$(x - 2)^2 \geq 0$ for every real x .

Step 2. Find the smallest possible output.

$$f(x) = (x - 2)^2 - 4 \geq -4.$$

Step 3. Show the bound is attained.

At $x = 2$, $f(2) = -4$. Thus the minimum *value* is -4 , not the minimizing input 2 .

Final answer: A; -4

2.3 / Question 54

Let the function p be defined as $p(x) = \frac{(x - c)^2 + 160}{2c}$, where c is a constant. If $p(c) = 10$, what is the value of $p(12)$?

A. 10.00

B. 10.25

C. 10.75

D. 11.00

Worked solution

2.3 / Q54

Step 1. Evaluate $p(c)$.

The denominator requires $c \neq 0$.

$$p(c) = \frac{(c - c)^2 + 160}{2c} = \frac{80}{c} = 10.$$

Step 2. Solve for the constant.

$80 = 10c$, so $c = 8$.

Step 3. Evaluate the requested input.

$$p(12) = \frac{(12 - 8)^2 + 160}{16} = \frac{176}{16} = 11.$$

Final answer: D; 11

Verification Notes

Answer-key comparison. All 117 items match the supplied key in exact value or intended option: 23 in Section 2.1, 40 in Section 2.2, and 54 in Section 2.3. The qualification for 2.3 / Q46 is explained below.

2.3 / Q46: the options are not equivalent as printed.

At $x = 1$, the four options give 200, 160, 128, and 102.5. Therefore they cannot all be equivalent forms of one function. Choice C is the intended answer because $128 \cdot 2^{x-1}$ displays $f(1) = 128$ as its coefficient. The solution preserves the printed question and explicitly distinguishes this structural answer from the inaccurate equivalence claim.

2.3 / Q32: area units.

The area of square Q is measured in *square inches*, not inches. The correct formula remains $(x + 44)^2$, choice A.

Exact values and decimal alternatives.

The supplied key sometimes lists shortened or rounded decimals alongside a fraction. These are not exact equalities. This packet uses exact answers:

Question	Exact value	Decimal expansion
2.1 / Q5	$4/225$	$0.017777\dots$
2.1 / Q21	$361/8$	45.125
2.2 / Q27	$29/3$	$9.666666\dots$

Checks used throughout.

Polynomial expansion and factorization, original-equation substitution, discriminants and repeated roots, domain restrictions, rejection of extraneous roots, vertex and intercept checks, and exponential growth-factor checks. Graph reading for 2.3 / Q49 is checked against a rational function consistent with the plotted curve.

Quick Answer Reference

2.1 Equivalent Expressions

Q1: C	Q2: B	Q3: $\frac{7}{6}$	Q4: C
Q5: $\frac{4}{225}$	Q6: -419	Q7: D	Q8: A
Q9: $0.09 = \frac{9}{100}$	Q10: B	Q11: D	Q12: D
Q13: D	Q14: D	Q15: D	Q16: C
Q17: 8	Q18: C	Q19: D	Q20: 6632
Q21: $\frac{361}{8} = 45.125$	Q22: C	Q23: C	

2.2 Nonlinear Equations and Systems

Q1: 9	Q2: B	Q3: 6	Q4: A
Q5: A	Q6: 5	Q7: D	Q8: $\frac{81}{4} = 20.25$
Q9: -3	Q10: A	Q11: 51	Q12: 5
Q13: A	Q14: 50	Q15: C	Q16: B
Q17: 2 or 8	Q18: 289	Q19: -1	Q20: D
Q21: D	Q22: D	Q23: D	Q24: C
Q25: D	Q26: A	Q27: $\frac{29}{3}$	Q28: A
Q29: A	Q30: C	Q31: C	Q32: 5
Q33: $\frac{1}{16} = 0.0625$	Q34: D	Q35: C	Q36: C
Q37: A	Q38: $\frac{29}{2} = 14.5$	Q39: C	Q40: -10

2.3 Nonlinear Functions

Q1: D	Q2: A	Q3: D	Q4: D
Q5: 20	Q6: D	Q7: -324	Q8: C
Q9: D	Q10: D	Q11: A	Q12: 76
Q13: B	Q14: 7	Q15: 5	Q16: D
Q17: B	Q18: A	Q19: B	Q20: C
Q21: A	Q22: D	Q23: D	Q24: D
Q25: B	Q26: B	Q27: 13	Q28: C
Q29: $\frac{86}{25} = 3.44$	Q30: A	Q31: B	Q32: A
Q33: B	Q34: C	Q35: -2112	Q36: C
Q37: B	Q38: 450	Q39: $\frac{25}{4} = 6.25$	Q40: 210
Q41: B	Q42: -7	Q43: D	Q44: D
Q45: D	Q46: C*	Q47: D	Q48: 8
Q49: C	Q50: D	Q51: C	Q52: C
Q53: A	Q54: D		

*2.3 / Q46: C is the intended answer; see Verification Notes for the source wording issue.